

HOSPITAL MANAGEMENT SYSTEM

Project submitted to the

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for the partial fulfillment of the requirements to award the degree of

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In

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USER VIEW ANALYSIS

Hospital Management System (HMS)

I. Introduction:

- The Hospital Management System (HMS) is designed to automate and manage hospital operations efficiently. The system follows a role-based user view architecture, where different users interact with the system based on their responsibilities.
- The three primary user views are:
 - Patient View
 - Doctor View
 - Admi View
- Each user has specific functionalities and controlled access to system data.

II. Patient View Analysis:

- *Overview:*
*The Patient View represents the **end-user (service consumer)** who interacts with the system to access healthcare services such as booking appointments and viewing medical records.*

➤ *Functional Modules:*

- ***Registration & Login:***

- ❖ Patients create accounts stored in the users table
- ❖ Authentication is handled through login credentials
- ❖ Login activity is tracked in userlog table

- ***Profile Management:***

- ❖ Patients can update:
 - ✓ Name, address, gender, city
- ❖ Data stored in:
 - ✓ users table

- ***Doctor Search & Selection:***

- ❖ Patients search doctors based on:
 - ✓ Specialization (from doctorspecilization)
 - ✓ Availability
- ❖ Doctor details fetched from:
 - ✓ doctors table
- ❖ Enables easy doctor discovery

- **Appointment Booking:**

- ❖ Patients book appointments which are stored in:

- ✓ appointment table

- ❖ Fields include:

- ✓ doctorId

- ✓ userId

- ✓ date & time

- ✓ status (userStatus, doctorStatus)

- ❖ This is the core functionality of the system

- **Appointment Tracking:**

- ❖ Patients can:

- ✓ View appointment status

- ✓ Check approval/rejection

- ❖ Status is controlled by both patient and doctor flags

- **Medical History Access:**

- ❖ Patients can view medical records stored in:

- ✓ Tblmedicalhistory

❖ Includes:

- ✓ Blood pressure
- ✓ Sugar level
- ✓ Prescriptions

❖ Ensures transparency and continuity of care

• **Aadhaar Health Integration:**

❖ Data stored in:

- ✓ aadhar_health table

❖ Features:

- ✓ Aadhaar linking
- ✓ Insurance status (SELF / PARENT / NONE)

❖ Provides identity verification and centralized health data

• **Contact Support:**

❖ Messages stored in:

- ✓ tblcontactus table

❖ Used for:

- ✓ Queries
- ✓ Complaints

➤ *Patient View Summary:*

- *User-friendly interface*
- *Self-service system*
- *Focus on booking and record access*

III. *Doctor View Analysis:*

➤ Overview:

- The Doctor View represents the service provider, responsible for patient treatment, appointment management, and medical record updates.

➤ Functional Modules:

- Login System:
 - ❖ Doctor credentials stored in:
 - ✓ doctors table
 - ❖ Login tracking in:
 - ✓ doctorslog table
- Profile Management:
 - ❖ Doctors update:
 - ✓ Specialization
 - ✓ Fees
 - ✓ Contact details

- Appointment Management:
 - ❖ Doctors view appointments from:
 - ✓ appointment table
 - ❖ Actions:
 - ✓ Accept or Reject appointment
 - ❖ Controlled by:
 - ✓ doctorStatus field
- Patient Information Access:
 - ❖ Doctors access patient details from:
 - ✓ Tblpatient
 - ✓ Users
 - ❖ Helps in diagnosis preparation
- Medical Record Management:
 - ❖ Doctors:
 - ✓ Add patient records
 - ✓ Update medical history
 - ❖ Stored in:
 - ✓ Tblmedicalhistory
 - ❖ Core medical functionality

- Doctor View Summary:
 - *Decision-making authority*
 - *Handles treatment lifecycle*
 - *Maintains medical data*

IV. Admin View Analysis:

- Overview:
 - The Admin View represents the system controller, responsible for managing system data, monitoring activities, and ensuring smooth operation.
- Functional Modules:
 - **Authentication:**
 - ❖ Admin credentials stored in:
 - ✓ admin table
 - **Doctor Management:**
 - ❖ Add / update / delete doctors
 - ❖ Data stored in:
 - ✓ Doctors
 - **Specialization Management:**

- ❖ Managed using:
 - ✓ Doctorspecilization

- ❖ Helps patient search functionality

- **Monitoring & Logs:**

- ❖ User activity:
 - ✓ Userlog

- ❖ Doctor activity:
 - ✓ Doctorslog

- ❖ Ensures security and accountability

- **Appointment Monitoring:**

- ❖ Admin can view all appointments from:
 - ✓ appointment table

- **Contact Message Handling:**

- ❖ Managed through:
 - ✓ Tblcontactus

- **CMS Management:**

- ❖ Website pages stored in:
 - ✓ tblpage

❖ Allows dynamic content updates

➤ Admin View Summary:

- *Full system control*
- *Data management*
- *Monitoring and maintenance*

V. Overall User Interaction Analysis:

➤ Interaction Flow:

- Patient registers and logs in
- Patient searches doctor and books appointment
- Appointment stored in database
- Doctor reviews and accepts/rejects
- Doctor updates medical history
- Admin monitors all activities

➤ Data Flow Between Users:

<i>FROM USER</i>	<i>TO USER</i>	<i>DATA</i>
<i>Patient</i>	<i>Doctor</i>	<i>Appointment</i>
<i>Doctor</i>	<i>Patient</i>	<i>Treatment/Records</i>
<i>Admin</i>	<i>All</i>	<i>Control & Monitoring</i>

VI. Key System Features Identified:

- Role-Based Access Control
- Appointment Management System
- Electronic Medical Records (EMR)
- Aadhaar Health Integration
- Logging & Monitoring System
- CMS & Support Module

VII. Limitations & Improvements:

➤ *Current Limitations:*

- *No payment integration*
- *No real-time notifications*
- *No emergency booking*

➤ *Suggested Enhancements:*

- *Online payment system*
- *SMS/Email notifications*
- *Doctor availability calendar*
- *Prescription download system]*

VIII. Conclusion:

➤ *The HMS system follows a **three-tier architecture**:*

- *Patient → Service Consumer*
- *Doctor → Service Provider*
- *Admin → System Controller*

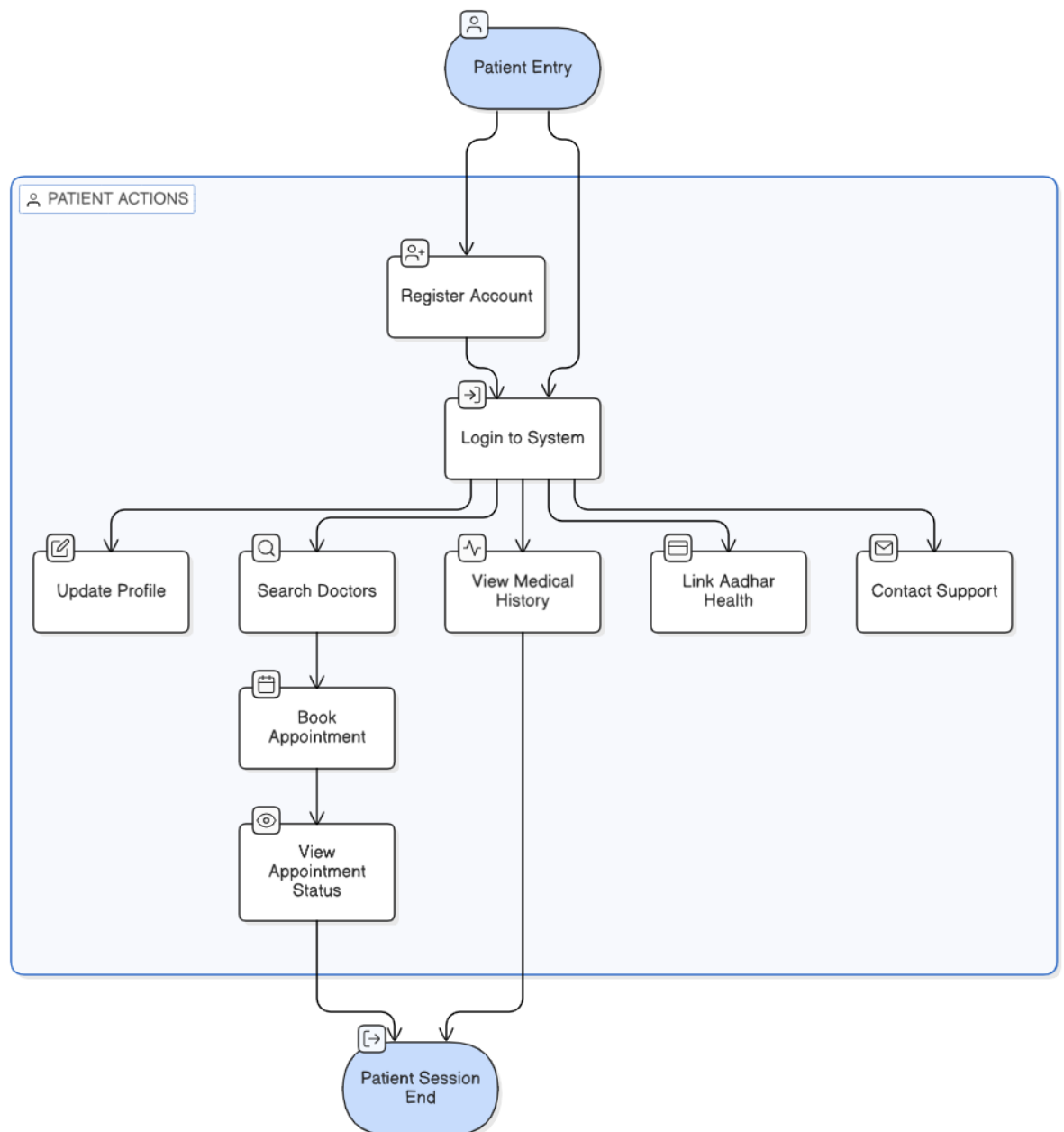
➤ *The system ensures:*

- *Efficient hospital operations*
- *Secure data handling*
- *Better patient-doctor interaction*

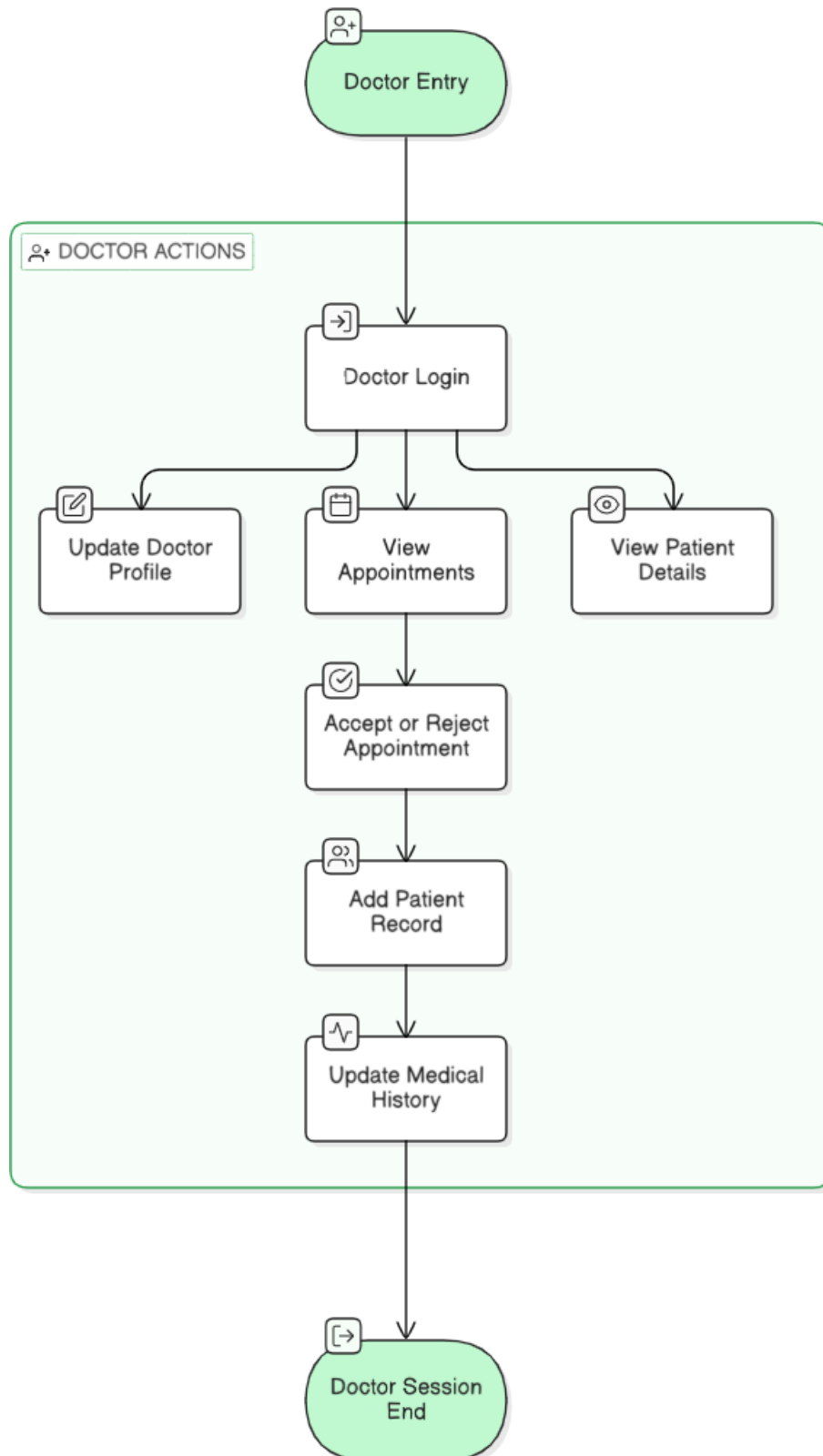
➤ *Overall, the design is **scalable, modular, and suitable for real-world implementation.***

USE CASE DIAGRAMS (UCD)

❖ USER PATIENT:



❖ USER DOCTOR:





USER ADMIN:

